CORE PURE / COMPLEX NUMBERS

Complex Numbers Basics

Questions by topic 13 questions · 94 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Instructions

CIE: this topic belongs to A Level Mathematics (9709), Paper 3 and is assumed knowledge for Further Mathematics 9231.

Questions

Question 1	*9 marks*	2
Question 2	*6 marks*	3
Question 3	*7 marks*	4
Question 4	*6 marks*	5
Question 5	*5 marks*	6
Question 6	*5 marks*	7
Question 7	*6 marks*	8
Question 8	*5 marks*	9
Question 9	*7 marks*	10
Question 10	*8 marks*	11
Question 11	*10 marks*	12
Question 12	*11 marks*	13
Question 13	*9 marks*	14

1. The complex numbers z and w are given by $z = 4 - 3i$ and $w = 1 + 2i$.

(a) Find, in the form $a + bi$, where a and b are real,

(i) $2z - w$, **[1]**

(ii) zw, **[2]**

(iii) $\frac{z}{w}$. **[3]**

(b) Find the real numbers p and q such that $pz + qw = 10 - 13i$. **[3]**

2. **(a)** Solve the equation

$$z^2 - 6z + 34 = 0$$

giving your answers in the form $a + bi$, where a and b are real. **[3]**

(b) Find the four roots of the equation

$$z^4 + 5z^2 - 36 = 0$$ **[3]**

3. **(a)** Let $z = x + iy$, where $x, y \in \mathbb{R}$.

(i) Write down z^*, the complex conjugate of z, in terms of x and y. **[1]**

(ii) Hence show that the product of a complex number and its conjugate is always real. **[2]**

(b) The complex number w satisfies

$$2w + 12 = 5w^* + 14i$$

(i) Find w, giving your answer in the form $a + bi$, where $a, b \in \mathbb{R}$. **[3]**

(ii) Calculate $w^2(w^*)^2$. **[1]**

4. The polynomial $f(z)$ is given by

$$f(z) = z^3 - 7z^2 + 17z - 15$$

(a) Show that $(z - 3)$ is a factor of $f(z)$. **[2]**

(b) Hence solve the equation $f(z) = 0$, giving the complex roots in the form $a+bi$, where a and b are real. **[4]**

5. The complex number z satisfies

$$4(z + 2i) = (1 - 2i)z^*$$

where z^* denotes the complex conjugate of z.
Determine z, giving your answer in the form $a + bi$, where a and b are real. **[5]**

6. Find the two complex numbers z that satisfy

$$z^2 = 21 - 20i$$

Give your answers in the form $a + bi$, where a and b are real. **[5]**

7. The cubic equation

$$z^3 + pz^2 + qz + 13 = 0$$

where p and q are real constants, has a root $2 - 3i$.

(a) Write down a second root of the equation. **[1]**

(b) Find the third root of the equation and the values of p and q. **[5]**

8. The complex numbers z_1 and z_2 are given by

$$z_1 = \sqrt{3}\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right)$$

and

$$z_2 = 2\left(\cos\left(\frac{5\pi}{6}\right) - i\sin\left(\frac{5\pi}{6}\right)\right)$$

(a) State the exact value of

(i) $|z_1 z_2|$,

(ii) the principal argument of $z_1 z_2$, taking $-\pi < \arg(z_1 z_2) \leq \pi$.

[2]

Let $w = z_1 z_2$.

(b) Determine

(i) the smallest positive integer n for which w^n is a positive real number,

(ii) the exact value of $|w^n|$ for this value of n.

[3]

9. In this question the argument of a complex number is taken in the range $0 \leq \arg z < 2\pi$.
Complex numbers z_1 and z_2 are defined by $z_1 = 4 + 3i$ and $z_2 = -12 + 5i$.

(a) Determine $z_1 z_2$, giving your answer in the form $a + bi$. **[2]**

(b) Express z_2 in modulus-argument form. **[2]**

(c) By considering the formula for $\tan(A - B)$, verify that $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$. **[3]**

10. The complex numbers z_1 and z_2 satisfy

$$z_1 = -3 + 2i$$
$$|z_1 z_2| = 26\sqrt{2}$$
$$\arg(z_1 z_2) = -\frac{\pi}{4}$$

(a) Express z_1 in the form $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$.
Give r exactly and θ in radians to 4 significant figures. **[2]**

(b) Determine z_2, giving your answer in the form $a + bi$, where a and b are integers. **[6]**

11. The complex number w is given by

$$w = -6\sqrt{3} - 6i$$

(a) (i) Find $|w|$. **[2]**

(ii) Find the principal argument of w, taking $-\pi < \arg(w) \leq \pi$. **[2]**

The complex numbers z_1 and z_2 are given by

$$z_1 = a - 3i, \qquad z_2 = 2(\cos\theta + i\sin\theta)$$

where a is a positive real constant and $-\pi < \theta \leq \pi$.

(b) Given that $z_1 z_2 = w$,

(i) find the exact value of a. **[3]**

(ii) find θ, giving your answer as an exact multiple of π. **[3]**

12. The complex numbers z and w are given by $z = 3 + i$ and $w = 2 + 4i$.

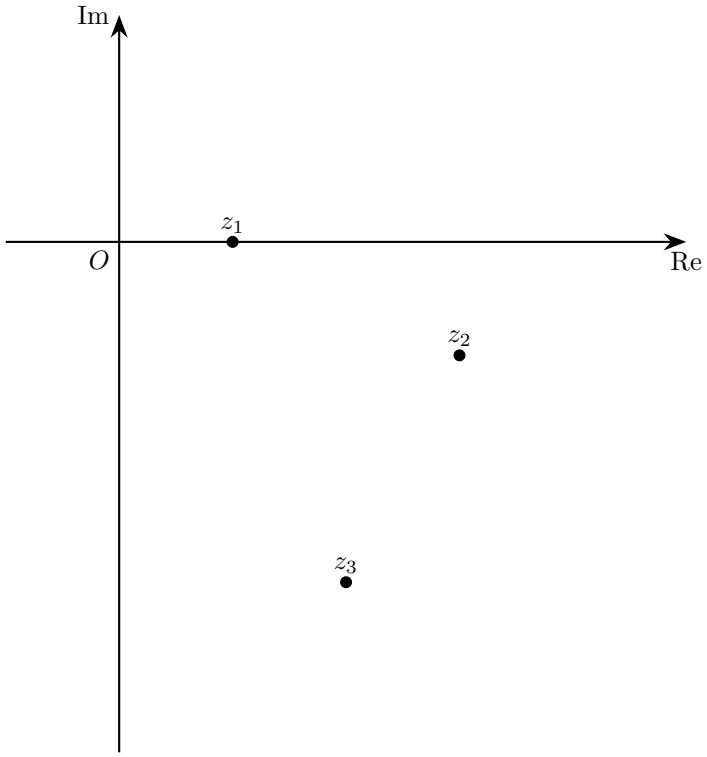
(a) (i) Find the modulus and principal argument of zw.
(ii) Express zw in the form $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$. **[4]**

(b) The complex number v satisfies

$$\frac{1}{v} = \frac{1}{w} - \frac{1}{z}$$

Find v in the form $a + bi$, where a and b are real. **[5]**

(c) The complex conjugate of v is denoted by v^*.
Show that $vv^* = k$, where k is a real number whose value is to be determined. **[2]**



13. The Argand diagram above shows the complex numbers $z_1 = 1$, $z_2 = 3 - i$ and $z_3 = 2 - 3i$.

(a) Explain why no polynomial equation of degree four with real coefficients can have all three of these numbers as roots. **[2]**

(b) Show that

$$\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \frac{\pi}{4}$$ **[3]**

(c) Hence show that

$$\arctan(3) - \arctan\left(\frac{1}{2}\right) = \frac{\pi}{4}$$ **[2]**

(d) Shade, on a copy of the diagram, the region consisting of all points whose complex coordinates satisfy

$$|z - 1| \leq |z - 2 + 3i|$$ **[2]**